

# hw1

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This is the link to the repo: <https://github.com/YifengHuang-Bob/Homework1>

## Problem 1 - Exploratory Analysis of Soybean Cultivars

**a.**

```
soybean <- read.csv("data/data.csv")

names(soybean) <- c(
  "Season", "Cultivar", "Repetition", "PH", "IFP", "NLP",
  "NGP", "NGL", "NS", "MHG", "GY"
)

class(soybean)
```

```
[1] "data.frame"
```

```
names(soybean)
```

```
[1] "Season"      "Cultivar"    "Repetition"  "PH"          "IFP"
[6] "NLP"         "NGP"         "NGL"         "NS"          "MHG"
[11] "GY"
```

**b.**

```
nrow(soybean)
```

```
[1] 320
```

```
length(unique(soybean$Cultivar))
```

```
[1] 40
```

```
length(unique(soybean$Season))
```

```
[1] 2
```

```
table(soybean$Season)
```

```
 1  2  
160 160
```

The dataset contains 320 observations, 40 distinct cultivars, and observations from 2 seasons. There are 160 observations from Season 1 and 160 observations from Season 2.

**c.**

**1.**

```
cor(soybean$PH, soybean$GY)
```

```
[1] 0.1232807
```

The correlation between plant height (PH) and grain yield (GY) is approximately 0.123.

**2.**

```
cor(  
  soybean$PH[soybean$Season == 1],  
  soybean$GY[soybean$Season == 1]  
)
```

```
[1] 0.2308581
```

```
cor(  
  soybean$PH[soybean$Season == 2],  
  soybean$GY[soybean$Season == 2]  
)
```

```
[1] -0.0227046
```

The correlation between PH and GY is approximately 0.231 for Season 1 and  $-0.023$  for Season 2. Therefore, Season 1 has the higher correlation.

3.

```
soybean$GY[which.max(soybean$PH)]
```

```
[1] 3304.678
```

The grain yield of the soybean observation with the greatest plant height is 3304.678.

4.

```
mean_mhg <- mean(soybean$MHG)
```

```
mean_mhg
```

```
[1] 168.3223
```

```
mean(soybean$MHG > mean_mhg) * 100
```

```
[1] 47.1875
```

47.1875% of the observations have an MHG greater than the overall average MHG.

5.

```
mean_gy_by_cultivar <- tapply(
  soybean$GY,
  soybean$Cultivar,
  mean
)

mean_gy_by_cultivar[which.max(mean_gy_by_cultivar)]
```

```
98R30 CE
3964.968
```

The cultivar with the highest average grain yield is 98R30 CE, with an average grain yield of 3964.968.

d.

```
quantitative_vars <- c(
  "PH", "IFP", "NLP", "NGP", "NGL", "NS", "MHG", "GY"
)

average_table <- rbind(
  Overall = colMeans(soybean[quantitative_vars]),
  `Season 1` = colMeans(
    soybean[soybean$Season == 1, quantitative_vars]
  ),
  `Season 2` = colMeans(
    soybean[soybean$Season == 2, quantitative_vars]
  )
)

round(average_table, 2)
```

|          | PH    | IFP   | NLP   | NGP    | NGL  | NS   | MHG    | GY      |
|----------|-------|-------|-------|--------|------|------|--------|---------|
| Overall  | 68.39 | 15.46 | 59.09 | 135.09 | 2.29 | 4.07 | 168.32 | 3418.55 |
| Season 1 | 65.99 | 16.48 | 57.97 | 135.24 | 2.30 | 4.82 | 162.29 | 3428.45 |
| Season 2 | 70.79 | 14.45 | 60.21 | 134.94 | 2.28 | 3.33 | 174.36 | 3408.66 |

e.

1.

Let  $\mu_1$  be the mean grain yield (GY) for Season 1 and  $\mu_2$  be the mean grain yield for Season 2.

$$H_0 : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2$$

2.

```
t.test(
  soybean$GY[soybean$Season == 1],
  soybean$GY[soybean$Season == 2]
)
```

Welch Two Sample t-test

```

data:  soybean$GY[soybean$Season == 1] and soybean$GY[soybean$Season == 2]
t = 0.35155, df = 231.85, p-value = 0.7255
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -91.15712 130.75170
sample estimates:
mean of x mean of y
 3428.452  3408.655

```

### Manual calculation

```

gy1 <- soybean$GY[soybean$Season == 1]
gy2 <- soybean$GY[soybean$Season == 2]

n1 <- length(gy1)
n2 <- length(gy2)

mean1 <- mean(gy1)
mean2 <- mean(gy2)

var1 <- var(gy1)
var2 <- var(gy2)

t_stat <- (mean1 - mean2) /
  sqrt(var1 / n1 + var2 / n2)

df <- (var1 / n1 + var2 / n2)^2 /
  ((var1 / n1)^2 / (n1 - 1) +
   (var2 / n2)^2 / (n2 - 1))

p_value <- 2 * pt(-abs(t_stat), df = df)

t_stat

```

```
[1] 0.351546
```

```
df
```

```
[1] 231.8466
```

```
p_value
```

```
[1] 0.725498
```

The mean grain yield is 3428.452 for Season 1 and 3408.655 for Season 2. The Welch two-sample t-test gives a p-value of 0.7255. Since the p-value is greater than 0.05, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the mean grain yield differs between the two seasons.

## Problem 2 - Adult Income Dataset

a.

```
adult <- read.csv(  
  "data/adult.data",  
  header = FALSE,  
  strip.white = TRUE,  
  na.strings = "?"  
)  
  
class(adult)
```

```
[1] "data.frame"
```

```
dim(adult)
```

```
[1] 32561    15
```

b.

```
names(adult) <- c(  
  "age", "workclass", "fnlwgt", "education", "education_num",  
  "marital_status", "occupation", "relationship", "race", "sex",  
  "capital_gain", "capital_loss", "hours_per_week",  
  "native_country", "income"  
)  
  
names(adult)
```

```
[1] "age"          "workclass"    "fnlwgt"      "education"
[5] "education_num" "marital_status" "occupation"   "relationship"
[9] "race"         "sex"          "capital_gain" "capital_loss"
[13] "hours_per_week" "native_country" "income"
```

c.

```
colSums(is.na(adult))
```

|                |              |                |                |               |
|----------------|--------------|----------------|----------------|---------------|
| age            | workclass    | fnlwgt         | education      | education_num |
| 0              | 1836         | 0              | 0              | 0             |
| marital_status | occupation   | relationship   | race           | sex           |
| 0              | 1843         | 0              | 0              | 0             |
| capital_gain   | capital_loss | hours_per_week | native_country | income        |
| 0              | 0            | 0              | 583            | 0             |

Missing values occur only in `workclass`, `occupation`, and `native_country`. There are 1836 missing observations for `workclass`, 1843 for `occupation`, and 583 for `native_country`. All other variables have no missing observations.

d.

```
nrow(adult)
```

```
[1] 32561
```

```
adult_filtered <- adult[adult$hours_per_week >= 10, ]
```

```
nrow(adult_filtered)
```

```
[1] 32103
```

The sample size is 32,561 before filtering and 32,103 after restricting the sample to individuals working at least 10 hours per week.

e.

```
nrow(adult_filtered)
```

```
[1] 32103
```

```
adult_complete <- adult_filtered[
  !is.na(adult_filtered$occupation) &
  !is.na(adult_filtered$workclass) &
  !is.na(adult_filtered$native_country),
]

nrow(adult_complete)
```

```
[1] 29873
```

```
colSums(
  is.na(
    adult_complete[,c("occupation", "workclass", "native_country")]
  )
)
```

| occupation | workclass | native_country |
|------------|-----------|----------------|
| 0          | 0         | 0              |

Observations with missing values in occupation, workclass, or native\_country were removed because these variables are needed for the analysis, and observations with missing values in these fields would have incomplete information.

f.

```
adult_clean <- adult_complete[
  adult_complete$age >= 18 &
  adult_complete$age <= 90 &
  adult_complete$education != "Preschool",
]

range(adult_clean$age)
```

```
[1] 18 90
```

```
sum(adult_clean$education == "Preschool")
```

```
[1] 0
```

**g.**

```
nrow(adult_clean)
```

```
[1] 29523
```

The final sample size is 29,523 observations.

## Problem 3 - Happy Numbers

**a.**

```
#' Compute the next value in a happy-number sequence
#'
#' Given a positive integer, this function computes the sum of the
#' squares of its digits.
#'
#' @param n A positive integer.
#' @return A positive integer equal to the sum of the squared digits of `n`.
#'
#' @examples
#' nextHappy(19)
#' nextHappy(82)

nextHappy <- function(n) {
  if (
    length(n) != 1 ||
    !is.numeric(n) ||
    is.na(n) ||
    !is.finite(n) ||
    n <= 0 ||
    n %% 1 != 0
  ) {
```

```

    stop("n must be a positive integer.")
  }

  digits <- as.numeric(strsplit(as.character(n), "")[[1]])

  sum(digits^2)
}

nextHappy(19)

```

```
[1] 82
```

```
nextHappy(82)
```

```
[1] 68
```

**b.**

```

#' Generate a happy-number sequence
#'
#' Starting from a positive integer, this function repeatedly applies
#' `nextHappy()` until the sequence reaches 1 or enters a cycle.
#'
#' @param n A positive integer.
#' @return A list containing the happy-number sequence and `isHappy = TRUE`
#' if the input is happy; otherwise, a list containing `isHappy = FALSE`.
#'
#' @examples
#' happySequence(19)
#' happySequence(4)

happySequence <- function(n) {
  if (
    length(n) != 1 ||
    !is.numeric(n) ||
    is.na(n) ||
    !is.finite(n) ||
    n <= 0 ||
    n %% 1 != 0
  ) {

```

```

    stop("n must be a positive integer.")
  }

  sequence <- n
  current <- n

  while (current != 1) {
    current <- nextHappy(current)

    if (current %in% sequence) {
      return(list(isHappy = FALSE))
    }

    sequence <- c(sequence, current)
  }

  list(
    sequence = sequence,
    isHappy = TRUE
  )
}

```

### Example 1

```
happySequence(19)
```

```

$sequence
[1] 19 82 68 100 1

$isHappy
[1] TRUE

```

### Example 2

```
happySequence(4)
```

```

$isHappy
[1] FALSE

```

**c.**

**(a)**

```
numbers <- 1:1000

happy_results <- lapply(numbers, happySequence)

is_happy <- sapply(
  happy_results,
  function(x) x$isHappy
)

sum(is_happy)
```

```
[1] 143
```

There are 143 happy numbers among the integers from 1 through 1000.

**(b)**

```
mean(is_happy) * 100
```

```
[1] 14.3
```

Happy numbers make up 14.3% of the integers from 1 through 1000.

**(c)**

```
happy_numbers <- numbers[is_happy]

sequence_lengths <- sapply(
  happy_results[is_happy],
  function(x) length(x$sequence)
)

max_length <- max(sequence_lengths)

happy_numbers[sequence_lengths == max_length]
```

```
[1] 356 365 379 397 469 496 536 563 635 649 653 694 739 793 899 937 946 964 973
[20] 989 998
```

```
max_length
```

```
[1] 7
```

The longest happy sequences have length 7. The starting values that produce these sequences are 356, 365, 379, 397, 469, 496, 536, 563, 635, 649, 653, 694, 739, 793, 899, 937, 946, 964, 973, 989, and 998.